Biochemistry Cume
October 16, 2015

The oxidation of malate by NAD^+ to form oxaloacetate is a highly endergonic reaction under standard conditions ($\Delta G^{o\prime}$ = 29 kJ/mol). The reaction proceeds readily under physiological conditions. (R=8.31 J/mol/K, T = 298 K).

a. (25 pts) Why?
b. (45 pts) Assume an $[NAD^+]/[NADH]$ ratio of 8, and give the lowest [malate]/[oxaloacetate] ratio at which oxaloacetate can be formed from malate.
c. (10 pts) The enzyme that catalyzes this reaction is called malate dehydrogenase. Name one metabolic pathway in aerobic organism that utilizes this enzyme.
d. (10 pts) Name and state the purpose for an anabolic pathway occurring in plants and bacteria that utilizes malate dehydrogenase, malate synthase and isocitrate lyase.
e. (10 pts) What is the consequence for animals of not having the pathway described in part d?